

# Paper 1B practice 1

Planck's constant from a handful of LEDs

IB Physics (2023 syllabus)

**with worked answers**

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Total: **20 marks**

$c = 3.00 \times 10^8 \text{ m s}^{-1}$ ;  $e = 1.60 \times 10^{-19} \text{ C}$ . Show your working.

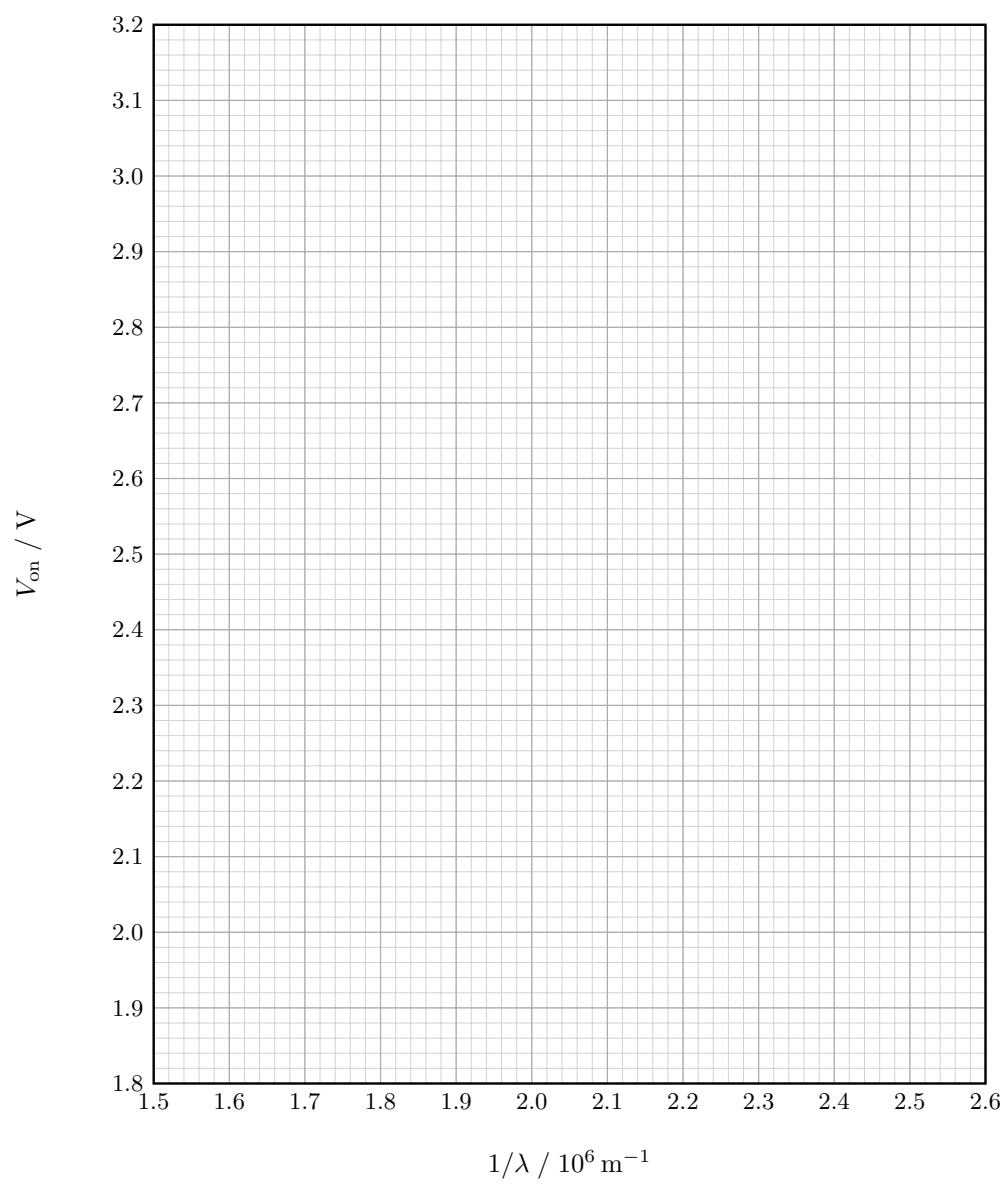
A light-emitting diode stays dark until the voltage across it reaches a turn-on value  $V_{\text{on}}$ . At that point an electron crossing the junction gives up its energy  $eV_{\text{on}}$  to create a single photon of the LED's wavelength  $\lambda$ , so

$$eV_{\text{on}} = \frac{hc}{\lambda}.$$

A student measures  $V_{\text{on}}$  for six LEDs of known wavelength. Each voltage is uncertain by  $\pm 0.05 \text{ V}$ .

| Colour | $\lambda / \text{nm}$ | $1/\lambda / 10^6 \text{ m}^{-1}$ | $V_{\text{on}} / \text{V}$ |
|--------|-----------------------|-----------------------------------|----------------------------|
| red    | 635                   | 1.575                             | 1.95                       |
| amber  | 605                   |                                   | 2.08                       |
| yellow | 590                   | 1.695                             | 2.10                       |
| green  | 525                   |                                   | 2.34                       |
| blue   | 470                   | 2.128                             | 2.67                       |
| violet | 400                   |                                   | 3.09                       |

- (a) Complete the third column of the table. [2]
- (b) Explain why a graph of  $V_{\text{on}}$  against  $1/\lambda$  should be a straight line through the origin, and state what its gradient represents. [3]
- (c) Plot the six points on the grid on the next page, and add error bars for the uncertainty in  $V_{\text{on}}$ . [3]
- (d) Draw the line of best fit. Then draw the steepest and the shallowest lines that still pass through all the error bars. [2]
- (e) Determine the gradient of your line of best fit, with its unit. [2]
- (f) Hence determine a value for the Planck constant, and use your steepest and shallowest lines to give its uncertainty. [4]
- (g) The accepted value is  $6.63 \times 10^{-34} \text{ J s}$ . Discuss whether this experiment is consistent with it. [2]
- (h) The student's best-fit line does not pass exactly through the origin, but cuts the voltage axis slightly above it. Suggest one physical reason. [2]



## Answers

- (a)  $1/\lambda$  in  $10^6 \text{ m}^{-1}$  is  $1000/(\lambda/\text{nm})$ : amber  $1000/605 = 1.653$ ; green  $1000/525 = 1.905$ ; violet  $1000/400 = 2.500$ .
- (b) Rearranging,  $V_{\text{on}} = \frac{hc}{e} \cdot \frac{1}{\lambda}$ . This has the form  $y = mx$  with no constant term, so plotting  $V_{\text{on}}$  against  $1/\lambda$  gives a straight line through the origin. The gradient is  $\frac{hc}{e}$ , and because  $h$ ,  $c$  and  $e$  are all constants the gradient is a fixed quantity from which  $h$  can be found.
- (c) Points at (1.575, 1.95), (1.653, 2.08), (1.695, 2.10), (1.905, 2.34), (2.128, 2.67), (2.500, 3.09), each with a vertical bar reaching 0.05 V above and below, which is half a major division on this scale.
- (d) A single straight line passing through or close to every error bar, with roughly equal numbers of points above and below. The steepest and shallowest lines are most easily drawn from the bottom of the first error bar to the top of the last, and from the top of the first to the bottom of the last.
- (e) Using the extreme points of the best-fit line, for example (1.575, 1.96) to (2.500, 3.10):

$$\text{gradient} = \frac{3.10 - 1.96}{2.500 - 1.575} = 1.23 \text{ V per } 10^6 \text{ m}^{-1} = 1.23 \times 10^{-6} \text{ V m}.$$

Accept 1.20 to 1.27.

- (f) Since the gradient equals  $hc/e$ ,

$$h = \frac{e \times \text{gradient}}{c} = \frac{(1.60 \times 10^{-19})(1.23 \times 10^{-6})}{3.00 \times 10^8} = 6.6 \times 10^{-34} \text{ J s}.$$

The steepest and shallowest lines give gradients of about 1.34 and 1.12 (in  $10^{-6} \text{ V m}$ ), so  $h$  ranges from about  $6.0 \times 10^{-34}$  to  $7.1 \times 10^{-34} \text{ J s}$ . Half that spread is the uncertainty:  $h = (6.6 \pm 0.6) \times 10^{-34} \text{ J s}$ .

- (g) The accepted  $6.63 \times 10^{-34} \text{ J s}$  lies comfortably inside the range 6.0 to  $7.1 \times 10^{-34} \text{ J s}$ , so the experiment *is* consistent with the accepted value. The agreement is not impressive, though: the uncertainty is about 9% of the result, which is large because the turn-on voltage is hard to judge. The experiment supports the accepted value rather than testing it severely.
- (h) An LED does not switch on abruptly at exactly  $hc/\lambda e$ . Any of the following earns the marks: the current rises gradually, so the voltage judged as “turn-on” is always a little above the true threshold; there is a resistive voltage drop across the leads and the semiconductor itself; the photon emitted is not always of the peak wavelength, since an LED emits over a band. Each of these shifts every reading upwards by a similar amount, which is exactly what produces a small positive intercept rather than a change of gradient.